Induced Almost-Cliques and Edge Extremality in Triple Systems

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Abstract

Falgas–Ravry and Vaughan proved that the asymptotic maximum proportion of five-vertex sets spanning nine edges in a triple system is 5/8. We determine the exact maximum for every sufficiently large order and show that the balanced complete bipartite construction is the unique extremizer. The proof combines a rational flag certificate, induced removal, and local corrections at vertices, pairs, and triples. Consequently, for every sufficiently large odd order, no $K_5^{(3)}$-free triple system simultaneously maximizes its edge count and its number of induced five-vertex almost-cliques. This answers their question on simultaneous extremality negatively for a fixed complete forbidden graph. For every even $t \geq 6$, we also improve the induced $K_t^{(3)} - e$ density predicted by their balanced construction. Reweighting that same template gives an explicit strict improvement, although its ordinary edge density decreases.

1 Introduction

Edge and induced-subgraph extremal problems can favor different structures under the same forbidden-subgraph condition. Falgas–Ravry and Vaughan [3] studied their relation for triple systems. We determine the exact extremizers for induced five-vertex almost-cliques and use their structure to separate the two objectives.

All hypergraphs in this paper are finite and simple. A 3-*graph*, or triple system, has edges of size three. Write $K_t^{(3)}$ for the complete 3-graph on t vertices, and put $F = K_5^{(3)} - e$, where one edge is deleted. For a 3-graph H, let $I(H)$ be the number of five-element subsets of $V(H)$ that induce F. Thus each copy is counted by its vertex set, with no labeling factor. We write $e(H) = |E(H)|$.

For disjoint sets A, B, let $\mathcal{B}[A, B]$ contain precisely the triples meeting both parts. Let $\mathcal{B}_n$ denote this construction with part sizes $a = \lfloor n/2 \rfloor$ and $b = \lceil n/2 \rceil$. A five-set induces F exactly when it meets the two parts in three and two vertices, so

$$I(\mathcal{B}[A, B]) = \binom{|A|}{3}\binom{|B|}{2} + \binom{|A|}{2}\binom{|B|}{3} = \frac{n-4}{12}|A||B|(|A||B| - n + 1), \tag{1}$$

where $n = |A| + |B|$. In particular, $I(\mathcal{B}_n) = (1/192 + o(1))n^5$.

Falgas–Ravry and Vaughan [3, Theorem 34] proved that the maximum of $I(H)/\binom{n}{5}$ tends to 5/8. Immediately after that theorem, they announced that $\mathcal{B}_n$ is the stable extremum and referred to a forthcoming manuscript. Our first result is an exact finite statement, including uniqueness. The proof below supplies the stability input directly rather than assuming the announced result.

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Theorem 1.1. *There exists N such that every 3-graph H on $n \geq N$ vertices satisfies*

$$I(H) \leq \binom{\lfloor n/2 \rfloor}{3}\binom{\lceil n/2 \rceil}{2} + \binom{\lfloor n/2 \rfloor}{2}\binom{\lceil n/2 \rceil}{3}.$$

Equality holds if and only if H is isomorphic to $\mathcal{B}_n$.

Taking complements gives the corresponding theorem for five-sets spanning exactly one edge. The unique extremizer for that problem is the disjoint union of two complete 3-graphs whose orders differ by at most one.

Falgas–Ravry and Vaughan [3, Question 21] asked whether every order admits a simultaneous edge and induced-count extremizer, emphasizing the case of a complete forbidden graph and an almost-complete target. Our exact theorem gives a negative answer with the fixed forbidden graph $K_5^{(3)}$.

Theorem 1.2. *For every sufficiently large odd integer n, no $K_5^{(3)}$-free 3-graph on n vertices simultaneously maximizes $e(H)$ and $I(H)$ over all $K_5^{(3)}$-free 3-graphs of that order. The same conclusion holds for $n = 9$.*

The graph $\mathcal{B}_n$ is $K_5^{(3)}$-free and uniquely maximizes the induced count, but classical constructions have more edges at odd orders. We use the quantitative construction of Bluskov, de Heer, and Sidorenko [1]. A further comparison in Section 4 follows from the total-clique theorem of Chen, Deng, Hou, Liu, and Zhang [2].

A second question concerns the prescribed constructions for larger complete forbidden graphs. In Conjecture 5, Falgas–Ravry and Vaughan [3] proposed that a balanced construction, denoted $G_t(n)$ and attributed there to Keevash and Mubayi, solves both the ordinary and induced problems. We show that its induced-density prediction fails for every even $t \geq 6$.

Write $I_t(H)$ for the number of induced copies of $K_t^{(3)} - e$ and

$$\pi_t = \lim_{n\to\infty} \max_{\substack{|V(H)|=n \\ K_t^{(3)} \not\subseteq H}} \frac{I_t(H)}{\binom{n}{t}}.$$

The limit exists by averaging over vertex subsets. In particular, $I_5 = I$. The next result uses only a change of class proportions in the existing construction.

Theorem 1.3. *Let $t = 2s + 4$ with $s \geq 1$, and define*

$$p_t(b) = \frac{t!}{162 \cdot 2^s s^{2s}} b^3(5-b)(1-b)^{2s}, \qquad b_* = \frac{5t - 1 - \sqrt{(5t-1)^2 - 60t}}{2t}. \tag{2}$$

Then $0 < b_* < 3/(t-1)$ and

$$\pi_t \geq p_t(b_*) > p_t\left(\frac{3}{t-1}\right) = \frac{t!}{(t-1)^t}\frac{5t-8}{3}\,2^{(t-6)/2}.$$

The expression on the right is the limiting induced density of $G_t(n)$. In particular,

$$\pi_6 \geq \frac{24500000}{72412707} > \frac{1056}{3125}.$$

The finite universal formulation of Conjecture 5 was already incompatible with classical odd-order edge constructions; see Keevash [4, Section 9]. Theorem 1.3 concerns a different obstruction, namely a positive asymptotic improvement of the induced objective at every even target order. It does not refute the conjectured ordinary Turán density, and it does not determine π_t exactly.

The proof of Theorem 1.1 has two stages. Section 2 gives a rational flag inequality with positive slack on every six-vertex type that is not complete bipartite. Induced removal and a reconstruction argument then yield global stability. Section 3 turns this into exactness. A six-vertex colored graph inequality first controls every link; pair corrections then control all defect codegrees, after which any remaining incorrect triple can be changed profitably. Local modification arguments near the bipartite construction also appear in [2], for different objectives and under a $K_5^{(3)}$-free hypothesis. Our corrections are unrestricted, as required for Theorem 1.1.

2 Global stability

We first show that a 3-graph with nearly the maximum induced count differs from $\mathcal{B}_n$ in few triples. The computational ingredient is a finite rational inequality. We specify its normalization and verification before giving the passage from six vertices to arbitrary orders.

Let $\mathcal{C}_6$ be the four isomorphism types $\mathcal{B}[A,B]$ on six vertices, with $0 \le |A| \le 3$. For a 3-graph H on $n \ge 6$ vertices, let $p(H) = I(H)/\binom{n}{5}$, and let $b(H)$ be the proportion of six-sets inducing a graph outside $\mathcal{C}_6$.

Lemma 2.1. *There is an absolute constant C such that every 3-graph H on $n \ge 6$ vertices satisfies*

$$p(H) + \frac{1}{20} b(H) \le \frac{5}{8} + \frac{C}{n}. \tag{3}$$

Proof. We use seven positive semidefinite flag forms in the sense of Razborov [5]. The finite definitions below suffice for the argument. Label a six-vertex graph G by $0,\dots,5$ and put $E_{ijk} = \mathbf{1}_{\{i,j,k\}\in E(G)}$. Each matrix entry is the probability of an ordered pair of flag states under a uniformly random permutation of these six labels. All entries therefore have denominator 720.

The first block has no roots. Its two flags are the empty and complete triple, on positions $0,1,2$ and $3,4,5$. The second block has ordered roots $0,1$ and flags on four vertices. Encode a flag by

$$E_{012} + 2E_{013} + 4E_{023} + 8E_{123},$$

and identify the two codes obtained by swapping the two nonroot vertices. There are twelve flags. Their canonical codes, in order, are $0,1,3,4,5,7,8,9,11,12,13,15$. The two extensions use positions $2,3$ and $4,5$.

For each $j = 0,\dots,4$, one further block has ordered roots $0,1,2,3$, with root edge mask $2^j - 1$ on the ordered triples $012, 013, 023, 123$. Its 64 flags are encoded by the six incidences between the extra vertex and the root pairs, in lexicographic order. The two extensions use vertices 4 and 5. A contribution is zero unless the roots have exactly the specified labeled edge mask. In particular, no conditional renormalization by the probability of that root type is made.

Write $M_i(G)$ for the seven matrices just defined, indexed by $i = 0,\dots,6$. Their dimensions are $2, 12, 64, 64, 64, 64, 64$. The supplied integer matrices R_i, T_i define

$$Q_i = \frac{R_i T_i R_i^{\mathsf{T}}}{900000000}.$$

The matrices T_i are symmetric positive definite, of dimensions $1, 10, 63, 64, 64, 63, 61$, respectively. Exact verification gives, for every six-vertex 3-graph G,

$$p(G) + \frac{1}{100}\mathbf{1}_{K_5^{(3)} \subseteq G} + \sum_{i=0}^{6} \langle M_i(G), Q_i \rangle \leq \frac{5}{8}, \tag{4}$$

where $\langle M, Q \rangle = \sum_{a,b} M_{ab} Q_{ab}$ includes both ordered off-diagonal entries. Equality holds precisely for the four graphs in $\mathcal{C}_6$. Every other graph has slack at least

$$\frac{278519687}{4500000000} > \frac{1}{20}.$$

Thus adding $\mathbf{1}_{G \notin \mathcal{C}_6}/20$ to the left side of (4) preserves the inequality.

The integer arrays are supplied in the supplementary file `certificate.json`. The lexicographically ordered triples of $\{0, \ldots, 5\}$ encode a graph by a 20-bit integer. The verifier checks all 2136 isomorphism representatives and obtains the four equality masks $0, 1023, 65535, 524286$. Fraction-free elimination checks that every leading principal minor of each T_i is positive, proving positive definiteness by Sylvester's criterion. It then reconstructs every Q_i and checks (4) using integers. A separate implementation regenerates all matrix coefficients from the sampling definition and verifies that the representative orbits partition all 2^{20} labeled graphs. The computation is finite and exact; no numerical optimization assumption enters the inequality.

It remains to justify averaging the matrix terms. Fix an ordered root embedding into an n-vertex graph, and choose the two extensions independently. If the roots have the required type, let v be the probability vector of a single extension. The expected quadratic form is $v^{\mathsf{T}} Q_i v \geq 0$; if the roots have a different type, it is zero. Conditioning the two extensions to be disjoint gives the matrix M_i averaged over random six-sets. Since at most three vertices are used in each extension, the probability of an intersection is $O(1/n)$. The entries of the seven fixed matrices are bounded, so the change in expectation is $O(1/n)$, uniformly in the graph. Enlarging the constant handles the finitely many smaller orders.

Finally, the average of $p(G)$ over six-subsets is exactly $p(H)$. Average the strengthened form of (4), discard its nonnegative complete-five-set term, and use the preceding lower bound on the flag contribution. This gives (3). □

To interpret the equality types globally, a complete four-set provides a consistent bipartition. No classification of graphs without such a four-set is needed.

Lemma 2.2. *Let G be a 3-graph on at least six vertices. If every induced six-vertex subgraph belongs to $\mathcal{C}_6$ and G contains $K_4^{(3)}$, then $G = \mathcal{B}[A, B]$ for some partition A, B.*

Proof. Fix a complete four-set R. For every $v \notin R$, extend $R \cup \{v\}$ to a six-set. In its bipartition, the four vertices of R occupy two positions in each part. Consequently, $R \cup \{v\}$ has a unique missing triple, of the form $P_v \cup \{v\}$ for a pair $P_v \subset R$.

For distinct $u, v \notin R$, the bipartition of $G[R \cup \{u, v\}]$ shows that either $P_u = P_v$ or $P_u = R \setminus P_v$. Choose one outside vertex v_0 and use the root pairs $P_{v_0}, R \setminus P_{v_0}$ to define two parts. Assign each outside vertex v to the part containing P_v. The preceding alternative makes this assignment consistent. The graphs on $R \cup \{v\}$ and $R \cup \{u, v\}$ show that every triple meeting R has its required status in $\mathcal{B}[A, B]$.

Consider three vertices u, v, w outside R. Choose the part containing at least two of them, and let a, b be its two root vertices. Let c be a root vertex in the other part. In a bipartition of the six-set $\{a, b, c, u, v, w\}$, every already absent triple abx, with x assigned to the first part, forces a, b, x

into one class. The edge abc forces c into the other class. If the remaining outside vertex is assigned to that other part, its triple with a, b is an edge and forces the same assignment. Thus uvw is an edge exactly when it meets both assigned parts. All triples now agree with $\mathcal{B}[A, B]$. □

Proposition 2.3. *For every $\varepsilon > 0$ there exist $\delta > 0$ and N such that every 3-graph H on $n \geq N$ vertices with*

$$I(H) \geq \left(\frac{1}{192} - \delta\right) n^5$$

can be changed into a copy of $\mathcal{B}_n$ by adding and deleting at most εn^3 edges.

Proof. It suffices to prove the sequence formulation. Let H range over a sequence with $n \to \infty$ and $I(H) \geq (1/192 - o(1))n^5$. Then $p(H) \geq 5/8 - o(1)$, so Lemma 2.1 gives $b(H) = o(1)$. Apply the induced removal lemma of Rödl and Schacht [6, Theorem 6] to the finite family of six-vertex 3-graphs outside $\mathcal{C}_6$. By changing $o(n^3)$ triples, obtain a graph G in which every six-set induces a member of $\mathcal{C}_6$.

A changed triple lies in exactly $\binom{n-3}{2}$ five-sets. Hence

$$|I(G) - I(H)| \leq |E(G) \triangle E(H)| \binom{n-3}{2} = o(n^5).$$

In particular, $I(G) > 0$ for large n. A five-set with nine edges contains three complete four-sets, so G contains $K_4^{(3)}$. Lemma 2.2 yields $G = \mathcal{B}[A, B]$.

Put $x = |A|/n$. Equation (1) gives, uniformly in the part sizes,

$$\frac{I(G)}{n^5} = \frac{x^2(1-x)^2}{12} + O(1/n).$$

The expression on the right is uniquely maximized at $x = 1/2$, where its value is $1/192$. Thus $|A|, |B| = n/2 + o(n)$. Moving $o(n)$ vertices balances the partition and changes only $o(n^3)$ further triples. This proves the sequence statement. If the quantified proposition failed, a fixed $\varepsilon > 0$ and successive choices $\delta \to 0$, $N \to \infty$ would give a sequence contradicting it. □

3 Exact extremality

Global stability alone permits defects concentrated around a vertex or a pair. We eliminate that possibility before changing individual triples. The following ordinary graph inequality controls the link of one vertex after the rest of the hypergraph has been made bipartite.

Lemma 3.1. *Let L be a graph with a specified partition into two sets of size three. Let $D(L)$ count four-sets meeting each part twice and inducing a diamond, that is, K_4 minus one edge. Let $T(L)$ count complete four-sets meeting the parts in three and one vertices. Then*

$$D(L) + 2T(L) \leq 15. \tag{5}$$

Equality holds exactly when all cross edges are present, one part induces a triangle, and the other is independent.

Proof. Let the two internal edge counts be $p, q \in \{0, 1, 2, 3\}$. Classifying a balanced four-set by the position of its missing edge bounds its contribution by $(3-p)q$, $p(3-q)$, or pq, according as that edge is in the first part, in the second, or crossing. Therefore

$$D(L) \leq 3(p+q) - pq, \qquad T(L) \leq 3(\mathbf{1}_{p=3} + \mathbf{1}_{q=3}).$$

If $p, q \le 2$, these bounds give a strict inequality in (5). If $p = 3$ and $q \le 2$, they give at most 15. Equality forces $T(L) = 3$, which requires every cross edge. Then $D(L) = 3(3 - q)$, so equality requires $q = 0$. Interchanging the two parts gives the other equality graph.

Suppose $p = q = 3$. Write the cross edges as a 3×3 zero-one matrix. The quantity $D(L)$ counts its 2×2 submatrices with exactly one zero. The quantity $T(L)$ is the number $R + C$ of full rows and full columns. If $R + C \le 2$, the bound $D(L) \le 9$ gives at most 13. If $R + C = 3$, equality at 15 would require one zero in all nine submatrices. Each zero belongs to four submatrices, making this impossible. If $R + C = 4$, there are two full rows and two full columns; the last entry is zero, and $D(L) + 2T(L) = 4 + 8 = 12$. The value $R + C = 5$ is impossible, and $R + C = 6$ gives $D(L) = 0$ and total 12. These cases prove the claim. □

Call either equality graph in Lemma 3.1 a *split graph for the specified partition.* In particular, this use of "split" requires all cross edges.

Lemma 3.2. *Let L be a sequence of graphs with a partition A, B, where $|A| = |B| = m \to \infty$. Let X count induced diamonds with two vertices in each part, and let Y count complete four-sets with three vertices in one part. If*

$$X + Y \ge \left(\frac{5}{12} - o(1)\right) m^4,$$

then L differs from one of the two split graphs for A, B in $o(m^2)$ edges. The conclusion is uniform over all such graphs.

Proof. Choose independently a uniform three-subset of each part. Averaging (5) gives

$$\frac{36}{m^2(m-1)^2}\left(X + Y + \frac{Y}{m-2}\right) \le 15. \tag{6}$$

Indeed, a fixed balanced four-set is selected with probability $36/[m^2(m-1)^2]$, whereas a fixed four-set of the other kind is selected with probability $18/[m^2(m-1)(m-2)]$. The coefficient 2 in (5) gives (6).

By the hypothesis, the expectation tends to 15. Since the slack in Lemma 3.1 is a nonnegative integer, the proportion ρ of sampled graphs that are not split tends to zero. The density of missing cross edges is at most ρ, by choosing a random cross pair inside the sample.

Let p_A, q_A be the probabilities that a random triple in A is complete or empty, and define p_B, q_B in the same way. Independence of the chosen triples gives

$$p_A q_B + q_A p_B \ge 1 - \rho.$$

Since $p_B + q_B \le 1$, at least one of p_A, q_A is at least $1 - \rho$. Suppose it is p_A. Then $q_A \le \rho$ and the last inequality gives $q_B \ge 1 - 2\rho$. Sampling a pair from a random triple shows that the missing-edge density in A is at most ρ, and the edge density in B is at most 2ρ. Together with the crossing estimate, at most $3\rho m^2$ changes give the corresponding split graph. The other case exchanges the two parts. This bound also proves uniformity. □

Proof of Theorem 1.1. Suppose the theorem fails for infinitely many orders. Choose an extremizer H of I that is not isomorphic to $\mathcal{B}_n$ along a sequence $n \to \infty$. Proposition 2.3 shows that H differs from a balanced complete bipartite graph in $o(n^3)$ triples. Choose a partition A, B maximizing the number of crossing edges of H, and set

$$\mathcal{D} = E(H) \triangle E(\mathcal{B}[A, B]).$$

Call its members *defect triples*. For a vertex or pair S, let $d_{\mathcal{D}}(S)$ denote the number of defect triples containing S.

The maximum partition has at least $e(\mathcal{B}_n) - o(n^3)$ crossing edges, while $e(H) = e(\mathcal{B}_n) + o(n^3)$. Hence it has only $o(n^3)$ internal edges. Since every complete bipartite graph on n vertices has at most $e(\mathcal{B}_n)$ edges, there are also only $o(n^3)$ missing cross edges. Thus $|\mathcal{D}| = o(n^3)$. The identity $e(\mathcal{B}[A, B]) = |A||B|(n-2)/2$ then forces $|A|, |B| = n/2 + o(n)$.

We first strengthen this global estimate to one that holds at every vertex. Let $i(v)$ be the number of target five-sets containing v, and $i(u, v)$ the number containing both vertices. Replace u by a copy of v, making every triple containing both u, v absent. A five-set containing both vertices cannot be a target, because at least three of its triples are absent. The exact change in I is therefore

$$i(v) - i(u) - i(u, v).$$

Extremality and $i(u, v) \leq \binom{n-2}{3}$ show that the values $i(v)$ differ by $O(n^3)$. Their average is $5I(H)/n$, so, uniformly in v,

$$i(v) \geq \left(\frac{5}{192} - o(1)\right) n^4. \tag{7}$$

Fix v and keep its link L_v, whose edges are the pairs xy with $vxy \in E(H)$. Replace the rest of H by $\mathcal{B}[A \setminus \{v\}, B \setminus \{v\}]$. At most $|\mathcal{D}|n = o(n^4)$ target five-sets through v are affected. In this comparison graph, the remaining four vertices of a target meet the parts twice each and induce a diamond in L_v, or meet them in three and one vertices and induce a complete graph in L_v. Other distributions cannot give nine triples.

Trim the larger of $A \setminus \{v\}, B \setminus \{v\}$ to obtain equal sizes $m = n/2 + o(n)$. This discards $o(n^4)$ four-sets. Equation (7) gives $X + Y \geq (5/12 - o(1))m^4$, so Lemma 3.2 puts the trimmed link within $o(n^2)$ edges of a split graph. Restoring the removed vertices costs only $o(n^2)$ changes.

Its orientation must agree with the assigned part of v. If $v \in A$ but its link were close to the split graph complete in $A \setminus \{v\}$ and empty in B, moving v to B would gain

$$e(L_v[A \setminus \{v\}]) - e(L_v[B]) = \frac{n^2}{8} + o(n^2)$$

crossing hyperedges. This contradicts the choice of the partition. The case $v \in B$ reverses the names of the parts. All preceding errors are uniform in the chosen vertex, and hence

$$\max_v d_{\mathcal{D}}(v) = o(n^2). \tag{8}$$

We next prove a uniform bound at pairs. Fix $\{u, v\}$ and make all triples not containing that pair ideal, while retaining the original triples containing it. Consider correcting the latter triples to their values in $\mathcal{B}[A, B]$. Comparing the actual change with its value in the ideal environment costs at most a constant times

$$n\big(d_{\mathcal{D}}(u) + d_{\mathcal{D}}(v)\big) + |\mathcal{D}| = o(n^3).$$

To see this, a defect meeting the pair once lies in at most n relevant five-sets, while a defect disjoint from the pair determines only one. This comparison applies both before and after the correction, and is uniform in the pair by (8).

Suppose first that $u, v \in A$. Put $a = |A| - 2$ and $b = |B|$. Let x be the number of missing neighbors of the pair in B, and y its number of present neighbors in $A \setminus \{u, v\}$. In the ideal environment the target count through this pair is

$$P(x, y) = (a - y)\binom{b - x}{2} + yx(b - x) + \binom{b - x}{3}. \tag{9}$$

For the first term, the other vertices consist of an internal nonneighbor and two external neighbors. For the second, they consist of an internal neighbor, one external neighbor, and one external nonneighbor. In both cases there is exactly one missing triple. The third term chooses three external neighbors; their internal triple in B is then the unique missing edge. Every other class distribution has at least two missing triples. The corrected pair has $x = y = 0$.

Set $m = n/2$, $s = x/m$, and $t = y/m$. After division by m^3, the expression in (9) converges uniformly on a fixed compact neighborhood of $[0,1]^2$ to

$$f(s,t) = \frac{(1-t)(1-s)^2}{2} + ts(1-s) + \frac{(1-s)^3}{6}.$$

Every limit of feasible (s,t) belongs to $[0,1]^2$. The maximum there is uniquely $f(0,0) = 2/3$. Indeed, f is linear in t. At $t = 0$ it decreases strictly with s, whereas at $t = 1$ it is at most $1/4 + 1/6 = 5/12$. At $s = 0$ its value is $2/3 - t/2$. Consequently, for any fixed $\eta > 0$, compactness gives a positive gap from $2/3$ when $s + t \geq \eta$. If $x + y$ were a positive proportion of n, correcting the pair would thus gain a positive multiple of n^3, contradicting extremality after the uniform $o(n^3)$ comparison error. It follows that $x + y = o(n)$ uniformly over same-part pairs.

For $u \in A$ and $v \in B$, put $a = |A| - 1$, $b = |B| - 1$, and let x, y count the missing neighbors in the two parts. The ideal target count is

$$\binom{a-x}{2}(b-y) + (a-x)\binom{b-y}{2}.$$

Here the remaining vertices must meet the two parts in two and one vertices, and all three must be neighbors of the chosen pair. At scale m^3, the limit is

$$\frac{(1-s)^2(1-t) + (1-s)(1-t)^2}{2},$$

whose unique maximum on $[0,1]^2$ is 1 at $(0,0)$. The same compactness argument gives $x + y = o(n)$ uniformly. Together, the two cases yield

$$\max_{u \neq v} d_{\mathcal{D}}(\{u,v\}) = o(n). \tag{10}$$

Finally, suppose $\mathcal{D}$ is nonempty and correct one defect triple e. In an otherwise ideal graph, correcting an internal triple in A creates $\binom{|B|}{2} = n^2/8 + o(n^2)$ targets. Correcting a triple with two vertices in A and one in B creates

$$(|A| - 2)(|B| - 1) + \binom{|B| - 1}{2} = \frac{3n^2}{8} + o(n^2)$$

targets. The opposite orientations exchange A and B. No ideal target is destroyed. The possible ideal five-set edge counts are $0, 6, 9$, so reverting a single incorrect triple cannot lose a target except by interference from another defect.

Another defect meeting e in two vertices affects at most n five-sets containing e; a defect meeting it once affects only one. Thus the number of exceptional five-sets is at most

$$n \sum_{S \in \binom{e}{2}} d_{\mathcal{D}}(S) + \sum_{v \in e} d_{\mathcal{D}}(v) = o(n^2),$$

by (10) and (8). Disjoint triples do not fit in a five-set. Hence the actual gain differs from the strictly positive ideal gain by $o(n^2)$ and is positive for large n, a contradiction. Thus $\mathcal{D}$ is empty.

Every extremizer in the chosen sequence is therefore complete bipartite. By (1), its count is $(n-4)z(z-n+1)/12$, where $z = |A||B|$. On the feasible interval $n-1 \le z \le \lfloor n^2/4 \rfloor$, this expression is strictly increasing for $n > 4$; empty parts give zero. It is therefore maximized only when the two part sizes differ by at most one. This contradicts the choice of H and proves Theorem 1.1. □

4 Separation of the two extremal problems

Since $\mathcal{B}_n$ is $K_5^{(3)}$-free, Theorem 1.1 also identifies the unique induced extremizer in that class. We first compare its edge count with odd-order constructions and give a self-contained nine-vertex example. We then turn to larger targets and reweight the construction proposed for even orders.

Proof of Theorem 1.2 for sufficiently large odd orders. Write $n = 2m + 1$. Bluskov, de Heer, and Sidorenko [1, Corollary 4] construct, for odd $n \ge 29$, a system of at most

$$\left\lceil \frac{m(m-2)(2m+1)}{6} \right\rceil$$

triples meeting every five-set. Its complement is $K_5^{(3)}$-free. The complement of $\mathcal{B}_n$ has

$$\binom{m}{3} + \binom{m+1}{3} = \frac{m(m-1)(2m-1)}{6}$$

edges, and subtracting the unrounded expression above gives $m/2$. Since the first quantity is integral, taking the ceiling gives

$$\mathrm{ex}(n, K_5^{(3)}) \ge e(\mathcal{B}_n) + \left\lfloor \frac{m}{2} \right\rfloor > e(\mathcal{B}_n). \tag{11}$$

The construction bound is unconditional; no conjectured matching lower bound for covering systems is used. For n also beyond the threshold in Theorem 1.1, any maximizer of I must be $\mathcal{B}_n$, while (11) shows that $\mathcal{B}_n$ does not maximize the edge count. This proves the asserted separation. □

The following quantitative comparison follows from the total-clique theorem of Chen, Deng, Hou, Liu, and Zhang and an elementary double count.

Proposition 4.1. *For every sufficiently large n, each $K_5^{(3)}$-free 3-graph H satisfies*

$$3I(H) + (n-4)e(H) \le 3I(\mathcal{B}_n) + (n-4)e(\mathcal{B}_n), \tag{12}$$

with equality if and only if $H \cong \mathcal{B}_n$. Consequently, for every sufficiently large odd $n = 2m+1$, every edge-extremal $K_5^{(3)}$-free H satisfies

$$I(H) \le I(\mathcal{B}_n) - \left\lceil \frac{(n-4)\lfloor m/2 \rfloor}{3} \right\rceil.$$

Proof. Let $c_4(H)$ count complete four-sets. Theorem 1.5 of Chen, Deng, Hou, Liu, and Zhang [2] says that, for all sufficiently large n, $\mathcal{B}_n$ uniquely maximizes the total number of cliques among $K_5^{(3)}$-free 3-graphs. Such a graph has no cliques of size at least five, so the count is $1 + n + \binom{n}{2} + e(H) + c_4(H)$. Therefore

$$e(H) + c_4(H) \le e(\mathcal{B}_n) + c_4(\mathcal{B}_n), \tag{13}$$

with equality only for $\mathcal{B}_n$.

Count pairs (Q, S) where Q is a complete four-set, S is a five-set, and $Q \subset S$. Each Q has $n-4$ extensions, and each target S contains exactly three complete four-sets. Thus

$$3I(H) \le (n-4)c_4(H).$$

For $\mathcal{B}_n$, every complete four-set meets each part twice, and every one-vertex extension is a target. The last inequality is therefore an equality for $\mathcal{B}_n$. Multiplying (13) by $n-4$ proves (12), including its equality condition. If H maximizes the edge count and n is odd, insert (11) into (12) and use the integrality of $I(H)$ to obtain the final bound. □

The value $\mathrm{ex}(9, K_5^{(3)}) = 72$ and its affine-plane construction are classical; see [1]. The following elementary proof also determines the induced count for every edge-extremal graph without a classification theorem.

Proposition 4.2. *Every nine-vertex $K_5^{(3)}$-free 3-graph with the maximum number of edges has 72 edges and 72 induced copies of F. The graph $\mathcal{B}_9$ has 70 edges and 100 induced copies of F.*

Proof. Call a triple system a *covering system* if every five-set contains an edge. Complements of such systems are precisely the $K_5^{(3)}$-free graphs. A seven-vertex covering system has at least five edges. To prove this, suppose it had four, adding edges if necessary. Its transversal number is at least three, since a transversal of size at most two would leave an independent five-set. Every vertex has degree at most two; a vertex meeting three edges and one vertex of a remaining edge would otherwise form a transversal of size two.

Let k, s be the numbers of vertices of degrees two and one. Then $2k + s = 12$ and $k + s \le 7$, so $k \ge 5$. Form a loopless multigraph on the four edge labels, assigning one multiedge to each degree-two vertex. Its maximum degree is at most three. Every two multiedges intersect, since disjoint ones would give a transversal of size two. A pairwise-intersecting family of pairs is contained in a star or a triangle. The star has at most three multiedges, and the triangle at most $\lfloor 9/2 \rfloor = 4$, contradicting $k \ge 5$.

Deleting each vertex and averaging now shows that a covering system on eight vertices has at least eight edges, and one on nine vertices at least twelve. Each triple survives five deletions in the first calculation and six in the second. Suppose a nine-vertex covering system J has exactly twelve edges. Each vertex-deleted graph has at least eight edges, so every vertex has degree at most four. Their sum is 36, and hence all degrees are four. Each $J - x$ has eight edges. Applying the seven-vertex bound to its vertex deletions gives maximum degree three in $J - x$, and its degree sum 24 forces every degree to equal three. Therefore each pair of vertices of J has codegree one.

Any two edges of J meet in at most one vertex, so three edges have union of size at least six. Every five-set of J thus has one or two edges. If their respective numbers are N_1, N_2, then

$$N_1 + N_2 = \binom{9}{5} = 126, \qquad N_1 + 2N_2 = 12\binom{6}{2} = 180.$$

It follows that $N_1 = 72$. Thus every complement with 72 edges has exactly 72 target five-sets.

For existence, take the twelve affine lines in the plane of order three. Any five points contain a line. Otherwise, from each point the other four would occupy the four directions exactly once; in any fixed direction the five points would then form a 1-regular graph, which is impossible. The twelve lines therefore form a covering system, and their complement establishes the ordinary extremal value 72.

Finally, $\mathcal{B}_9$ has part sizes four and five. Its edge count is $\binom{9}{3} - \binom{4}{3} - \binom{5}{3} = 70$, and its induced count is $\binom{4}{3}\binom{5}{2} + \binom{4}{2}\binom{5}{3} = 100$. It is $K_5^{(3)}$-free, so no graph with the maximum edge count can also maximize the induced count. This proves the remaining case of Theorem 1.2. □

We now prove Theorem 1.3. The complement formulation makes both the forbidden-clique condition and the induced count transparent.

Proof of Theorem 1.3. Fix $t = 2s + 4$, $s \geq 1$. Partition the vertex set into a special block A and s ordinary blocks $B_1, \ldots, B_s$. Divide A into three classes A_0, A_1, A_2. Define a 3-graph J with no edges between distinct blocks. Every B_j is complete. Inside A, include all triples within one class and all triples with two vertices in A_i and one in A_{i+1}, with indices modulo three. Let G be the complement of J.

Every four vertices of A contain an edge of J. If three lie in one class, they form an edge. Otherwise the nonzero class sizes are either $2, 2$ or $2, 1, 1$, and the cyclic rule supplies an edge. Thus an independent set of J has at most three vertices in A and two in each B_j, for a total of at most $3 + 2s = t - 1$. Consequently G is $K_t^{(3)}$-free, for all class sizes.

For $0 < b < 1$, give A asymptotic proportion b, divide it equally among its three classes, and give each B_j proportion $c = (1 - b)/s$. These proportions are realized by integer sizes with rounding errors bounded independently of n. Within A, a random triple has no edge with limiting probability $5/9$. The contributions are $2/9$ from three different classes and $1/3$ from the absent cyclic directions. A random four-set has exactly one edge with limiting probability $16/27$. A class distribution $2, 1, 1$ contributes $4/9$, and the three directions of distribution $3, 1, 0$ in which only the internal triple is an edge contribute $4/27$.

An induced copy of $K_t^{(3)} - e$ in G is a t-set with exactly one edge in J. If the edge lies in A, deleting a vertex from it leaves an independent subset of $J[A]$, so that block contains at most four chosen vertices. Every other block contains at most two. Since $t = 4 + 2s$, all these bounds must be equalities. If the edge lies in a block B_j, that block contains exactly three vertices, A contains exactly three independent vertices, and every other block contains two. These two possibilities are disjoint and exhaustive. The limiting induced density is therefore

$$\frac{t!}{4!(2!)^s}\frac{16}{27}b^4c^{2s} + s\frac{t!}{3!3!(2!)^{s-1}}\frac{5}{9}b^3c^{2s+1} = p_t(b).$$

This proves $\pi_t \geq p_t(b)$ for every $b \in (0, 1)$.

The construction $G_t(n)$ has equal proportions on its $t - 1$ underlying label classes. In the notation above it corresponds to $b_0 = 3/(t - 1)$ and $c = 2/(t - 1)$; the original template includes all triples on each ordinary pair of label classes, so those two classes combine into one complete block. Logarithmic differentiation gives

$$\frac{p_t'(b)}{p_t(b)} = \frac{3}{b} - \frac{1}{5 - b} - \frac{t - 4}{1 - b} = \frac{tb^2 - (5t - 1)b + 15}{b(5 - b)(1 - b)}.$$

At $b = b_0$, the first and third terms cancel, leaving $-1/(5 - b_0) < 0$. The numerator is positive at zero and negative at one. Being an upward quadratic, it has exactly one zero in $(0, 1)$, namely b_*. Thus p_t increases up to b_* and decreases thereafter, with $b_* < b_0$. Substituting b_0 in (2) gives the displayed balanced value.

For $t = 6$, choose the rational proportion $b = 10/17$. Direct substitution gives

$$p_6(10/17) = \frac{24500000}{72412707}, \qquad p_6(10/17) - p_6(3/5) = \frac{94681408}{226289709375} > 0.$$

The strict improvement therefore requires no numerical optimization. This completes the proof. □

There is also a small finite witness. For $t = 6$, write $(a_0, a_1, a_2; h)$ for the four block sizes above. The numbers of independent triples and four-sets with one edge in $J[A]$ are, respectively,

$$I_3 = a_0a_1a_2 + \sum_{i=0}^{2} \binom{a_i}{2} a_{i-1}, \qquad I_4 = \sum_{i=0}^{2} \left(\binom{a_i}{2} a_{i-1}a_{i+1} + \binom{a_i}{3} a_{i-1} \right).$$

The same exhaustive classification gives $I_6(G) = I_4\binom{h}{2} + I_3\binom{h}{3}$. The balanced choice $(12, 12, 12; 24)$ has 18,361,728 induced targets, while $(11, 12, 12; 25)$ has 18,367,800, an increase of 6072. Both graphs are $K_6^{(3)}$-free on 60 vertices. The supplementary verifiers check these counts by integer arithmetic and independently by class occupancies.

The change of weights serves the induced objective. In the same family the ordinary edge density is

$$1 - \frac{4}{9}b^3 - \frac{(1-b)^3}{s^2},$$

which is uniquely maximized at b_0, with value $1 - 4/(t-1)^2$. Thus the two objectives already prefer different weights within this family. This observation alone does not rule out a different simultaneous extremizer for even t.

Theorem 1.1 gives no explicit threshold and does not settle all smaller orders. The separation in Theorem 1.2 concerns every sufficiently large odd vertex order for the fixed forbidden graph $K_5^{(3)}$. The quantitative gap in Proposition 4.1 has order n^2. In contrast, Theorem 1.3 gives a positive induced-density improvement for every even target order at least six, while leaving its exact optimum open.

Declaration of AI use

GPT-6.1 was used to assist with exploring proof strategies, checking calculations and proofs, preparing the computational verification, and improving the exposition. The author takes responsibility for the content of the paper.

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